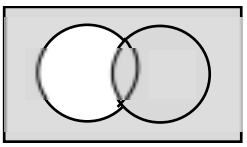
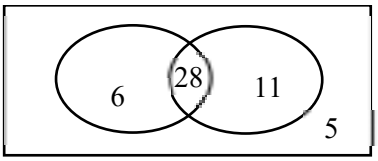
Question	Answer	Marks	Partial Marks
3(b)	953 or 952.6 to 952.8	**4**	**M3** for $\frac{1}{3}\times\pi\times7.6^2\times16\times\left(1-\left(\frac{16-12}{16}\right)^3\right)$ or $\frac{1}{3}\times\pi\times7.6^2\times16-\frac{1}{3}\times\pi\times1.9^2\times(16-12)$ OR **B1** for top radius = 1.9 or $\left(\frac{16-12}{16}\right)^3$ oe **M2** for $\frac{1}{3}\times\pi\times7.6^2\times16-\frac{1}{3}\times\pi\times(their\,1.9)^2\times(16-12)$ or $\frac{1}{3}\times\pi\times7.6^2\times16\times\left(1-their\left(\frac{16-12}{16}\right)^3\right)$ or **M1** for $\frac{1}{3}\times\pi\times7.6^2\times16$ or for $\frac{1}{3}\times\pi\times(their\,1.9)^2\times(16-12)$
4(a)(i)	438 cao	**2**	**M1** for $\frac{500}{1.142}$
4(a)(ii)	14.95	**2**	**M1** for [329 –] 275 × 1.142 oe
4(b)	14	**2**	**M1** for $5.25\times\frac{8}{3}$ oe
4(c)	1.7[0] or 1.699…	**3**	**M2** for $\sqrt[5]{\frac{6669}{6130}}$ or **M1** for 6669 = 6130 $(k)^5$
5(a)	13.5 or 13.47…	**4**	**B1** for angle 102 seen **M2** for $\sqrt{10.6^2+6.4^2-2\times10.6\times6.4\times\cos(180-78)}$ OR **M1** for $10.6^2+6.4^2-2\times10.6\times6.4\times\cos(180-78)$ **A1** for 181.5…
5(b)	8.68 or 8.682 to 8.683 nfww	**4**	**B1** for angle = 44 **M2** for sin(180 – 58 – 78) × $\frac{10.6}{\sin58}$ oe or **M1** for $\frac{\sin(180-58-78)}{x}=\frac{\sin58}{10.6}$ oe

Question	Answer	Marks	Partial Marks
5(c)	78.2 or 78.17 to 78.19…	**3**	**M2** for $\frac{1}{2}\times 10.6\times(6.4+\textit{their } 8.68)\times\sin(78)$ oe OR **M1** for $\frac{1}{2}\times 10.6\times 6.4\times\sin(180-78)$ oe **M1** for $\frac{1}{2}\times 10.6\times\textit{their } 8.68\times\sin 78$ oe
6(a)		**1**	
6(b)		**2**	**B1** for 2 or 3 correct elements or **M1** for $34-x$, x and $39-x$ correctly placed on diagram and $x = 28$
6(c)(i)	8	**1**	
6(c)(ii)	11	**1**	
6(c)(iii)	2	**1**	
6(c)(iv)	$C\cap S\cap B'$ oe	**1**	
6(c)(v)	$\frac{19}{30}$ oe	**1**	
6(c)(vi)	$\frac{2}{57}$ oe	**3**	**M2** for $\frac{4}{19}\times\frac{3}{18}$ or **M1** for $\frac{4}{19}$ seen
6(c)(vii)	Equal numbers 15 or equal probability $\frac{15}{30}$ oe	**1**	
7(a)	$\frac{x+5}{x+4}$ final answer	**3**	**B1** for $(x-5)(x+5)$ **B1** for $(x-5)(x+4)$
7(b)	$\frac{2x^2+12x-5}{x(x-1)}$ or $\frac{2x^2+12x-5}{x^2-x}$ final answer	**3**	**B1** for common denominator $x(x-1)$ oe **B1** for $(x-1)(x+5)+x(x+8)$ or better
7(c)(i)	$6x^2-8x$ final answer	**2**	**B1** for each term in final answer or **M1** for correct answer seen and spoilt